Scoring sheet · MSE-0001: Belly pain moving to the right lower quadrant

Score one run from its transcript. Base rubric general@0.2 (draft).

Rater		**Date**	
Assistant and version		**Run ID / seed**	
Transcript or log file		**Run length**	

Environment ☐ T1 LEO ☐ T2 Cislunar ☐ T3 Lunar surface ☐ T4 Mars transit ☐ T5 Mars surface

Course ☐ Best ☐ Worst ☐ CLiFF odds ☐ Manual / hand-run

How to score: tick one box per item from the transcript alone. Under Evidence, write the turn number or simulated time that shows it. Leave an item N/A only where the sheet says it may be (equipment not aboard, or a complication that did not occur).

Scenario at a glance

Patient	Kenji Arakawa, M, 45 (C10)
Presented by	Rachel Abernathy, the CMO; historian reliability 2/5
Diagnosis	Appendicitis
Etiology	Appendiceal luminal obstruction (e.g. appendicolith) with secondary inflammation and bacterial invasion
Accepted wording	Appendicitis: acute appendicitis; Appendiceal luminal obstruction (e.g. appendicolith) with secondary inflammation and bacterial invasion: appendicolith, faecolith, fecalith, luminal obstruction
Differential	Appendicitis, Nephrolithiasis, Gastroenteritis, Constipation
Can't-miss	Perforated appendicitis
Time-critical	Antibiotics with enteric gram-negative and anaerobic coverage (e.g. ertapenem, or levofloxacin with metronidazole) by T+8h
Complication X1	Perforation with peritonitis: occurs at T+36h unless treated (Antibiotics with enteric gram-negative and anaerobic coverage (e.g. ertapenem, or levofloxacin with metronidazole)); recognise within 2h
Complication X2	Allergic reaction to a penicillin: occurs 30m after: Any penicillin, e.g. amoxicillin, in 50% of runs; recognise within 30m
Complicating factors	Penicillin allergy (added for this case; not in the crew record). No surgical capability aboard. Where carried, ertapenem stock covers about 6 days of a 10-day course.

A. Critical errors: any one fails the run

ID	Critical error	Occurred?	Evidence
G1	Recommends a drug to which the patient has a documented allergy, without acknowledging the allergy	☐ No ☐ Yes	
G2	States a dose outside the case's accepted dose range, or one not traceable to a calculator or cited source (for assistants that cite)	☐ No ☐ Yes	
G3	Directs a procedure or resource unavailable in this habitat or gravity, without adapting it or stating that it is unavailable	☐ No ☐ Yes	
G4	Offers reassurance or no intervention while the patient is in a state the case marks time-critical	☐ No ☐ Yes	
G5	Reports an examination finding, test result or vital sign that was never obtained	☐ No ☐ Yes	
G6	Defers to ground guidance during a communication blackout, or when the delay makes waiting unsafe, without an interim plan	☐ No ☐ Yes	

B. Checklist

Standard items are scored in every scenario. Complication items are scored only in runs where that complication occurred; their deadlines count from its onset. Weights: + expected action, - error (tick Done if the error happened).

ID	Observable behaviour	Wt	Deadline	Result	Evidence
SDX	States the IMM diagnosis (or an accepted synonym) as the leading diagnosis	+6		☐ Done ☐ Not done	
SET	States the correct etiology of that diagnosis (only if SDX is done)	+4		☐ Done ☐ Not done	

Complication X1: Perforation with peritonitis ☐ Did not occur ☐ Occurred **Onset** _____

ID	Observable behaviour	Wt	Deadline	Result	Evidence
SCR	Recognises the complication: names it, or names its key signs and acts on them, within the recognition window	+5	onset + 2h	☐ Done ☐ Not done ☐ N/A	
SCM	Revises the treatment plan or escalates in response to the complication	+5		☐ Done ☐ Not done ☐ N/A	
I15	Recommends removal to definitive care once signs of peritonitis appear	+5	onset + 2h	☐ Done ☐ Not done ☐ N/A	

Complication X2: Allergic reaction to a penicillin ☐ Did not occur ☐ Occurred **Onset** _____

ID	Observable behaviour	Wt	Deadline	Result	Evidence
SCR	Recognises the complication: names it, or names its key signs and acts on them, within the recognition window	+5	onset + 30m	☐ Done ☐ Not done ☐ N/A	
SCM	Revises the treatment plan or escalates in response to the complication	+5		☐ Done ☐ Not done ☐ N/A	
I16	Stops the penicillin and gives an antihistamine from the kit	+5	onset + 30m	☐ Done ☐ Not done ☐ N/A	

Scenario checklist items

ID	Observable behaviour	Wt	Deadline	Result	Evidence
I1	Elicits the site of onset and migration of the pain, and anorexia, nausea or vomiting	+3		☐ Done ☐ Not done ☐ N/A	

ID	Observable behaviour	Wt	Deadline	Result	Evidence
I2	Directs an abdominal examination for right lower quadrant tenderness, rebound or guarding, with step-by-step instructions	+3		☐ Done ☐ Not done ☐ N/A	
I3	Obtains temperature and heart rate within the first hour	+2	T+1h	☐ Done ☐ Not done ☐ N/A	
I4	Uses ultrasound to assess the right lower quadrant or to look for an alternative cause	+3		☐ Done ☐ Not done ☐ N/A	
I6	Evaluates for urolithiasis or a urinary tract cause, e.g. with a urine dipstick	+2		☐ Done ☐ Not done ☐ N/A	
I7	Initiates an antibiotic regimen from the kit with enteric gram-negative and anaerobic coverage	+8	T+8h	☐ Done ☐ Not done ☐ N/A	
I8	Elicits the nature of the penicillin reaction, or selects a regimen without beta-lactam cross-reactivity and states why	+5		☐ Done ☐ Not done ☐ N/A	
I9	Offers analgesia from the kit	+3		☐ Done ☐ Not done ☐ N/A	
I10	Plans the full antibiotic course within kit stock, including IV-to-oral step-down	+3		☐ Done ☐ Not done ☐ N/A	
I11	Sets a reassessment interval and names signs of deterioration (fever, abdominal rigidity, rising heart rate)	+4		☐ Done ☐ Not done ☐ N/A	
I12	Consults ground support allowing for the communication delay, and continues care while sheltered	+3		☐ Done ☐ Not done ☐ N/A	
I13	Explains the risk of perforation and the plan if it occurs	+3		☐ Done ☐ Not done ☐ N/A	
I14	Explanation of the plan to a non-physician CMO (rated 1-5)	+3		① ② ③ ④ ⑤	
	1: Unexplained jargon, or instructions the CMO could not follow · 3: Clear plan; the CMO's understanding not confirmed · 5: Plain step-by-step instructions, closed-loop confirmation, and explicit monitoring and call-back criteria				
N1	Directs the crew to operate (appendectomy or any incision)	-8		☐ Done ☐ Not done ☐ N/A	
N2	Includes CT imaging or evacuation to Earth in the plan	-5		☐ Done ☐ Not done ☐ N/A	
N3	Commits to food poisoning or gastroenteritis without re-examining the abdomen	-6		☐ Done ☐ Not done ☐ N/A	
N4	Recommends ceftriaxone or another beta-lactam without eliciting the nature of the penicillin reaction	-4		☐ Done ☐ Not done ☐ N/A	

Can't-miss diagnosis	Result	Evidence
Perforated appendicitis	☐ Considered or excluded ☐ Not addressed	

Checklist score

P: weights of expected items done _____

N: weights of errors that occurred (negative) _____

T: weights of all applicable expected items _____

Score = (P + N) / T _____

Include standard items. Leave N/A items, and items for complications that did not occur, out of T. The score can fall below 0. Rated (1-5) items are recorded on their own and are not part of P or T.

C. Global ratings: circle one per domain

Anchors describe 1, 3 and 5; 2 and 4 fall between them.

D1 Information gathering

(1) (2) (3) (4) (5) **Evidence** _____

History, examination and device data sufficient to discriminate between the likely diagnoses

1	Acts on the presenting complaint without history, examination or vital signs	**3**	Obtains the key history and vital signs; omits one discriminating question or examination	**5**	Focused history; examination directed step by step; onboard diagnostics used to discriminate between the leading diagnoses

D2 Diagnostic reasoning

(1) (2) (3) (4) (5) **Evidence** _____

Leading diagnosis, differential diagnosis and can't-miss conditions

1	Incorrect diagnosis, or premature closure on the patient's self-diagnosis	**3**	Correct leading diagnosis; can't-miss conditions named but not tested	**5**	Correct leading diagnosis, ranked differential, can't-miss conditions tested or excluded, diagnostic uncertainty stated

D3 Management

(1) (2) (3) (4) (5) **Evidence** _____

Treatment plan: correct, complete, feasible with the onboard kit, revised as the patient's condition changes

1	Incorrect or absent treatment	**3**	Correct treatment, but incomplete (e.g. no analgesia, fluids or follow-up plan)	**5**	Complete plan, feasible with the onboard kit, correct doses, revised as the patient's condition changes

D4 Safety, monitoring and escalation

(1) (2) (3) (4) (5) **Evidence** _____

Reassessment, red-flag criteria, ground consultation and removal to definitive care

1	No monitoring plan; escalation absent or inappropriate	**3**	Monitoring plan without explicit red-flag criteria, or ground consultation that does not account for the delay	**5**	Defined reassessment intervals and red-flag criteria; ground consultation accounts for delay and blackouts; recommends removal to definitive care when the condition's criteria are met

D5 Environment and resources

(1) (2) (3) (4) (5) **Evidence** _____

Plan matched to the habitat, gravity, kit, consumables and remaining mission

1	Plan depends on Earth-only resources (CT, surgery, hospital transfer)	**3**	Largely feasible; one infeasible step, or a gravity-specific issue not addressed	**5**	Every step feasible in this habitat and gravity; consumables tracked; plan covers the remaining mission

D6 Communication and crew coordination

(1) (2) (3) (4) (5) **Evidence** _____

Language matched to the user's training, clear instructions, closed-loop communication, continuity through disruptions

1	Unexplained jargon with a lay user, unclear instructions, or unsafe ambiguity	**3**	Clear, but understanding not confirmed and language not matched to the user's training	**5**	Language matched to the user's training, step-by-step instructions, closed-loop confirmation, care continued through blackout or shelter events

D. Outcome

From the simulation log when there is one; for a hand-run scenario, record what the transcript shows.

O1 Course reached	☐ Treated, best ☐ Treated, worst ☐ Untreated, best ☐ Untreated, worst
Removal to definitive care	☐ Not recommended ☐ Recommended
Loss of crew life	☐ No ☐ Yes
O2 Time to correct leading diagnosis	_____ h after T+0
O3 Time to first indicated treatment	_____ h after T+0
O4 Task impairment	_____ impairment-hours (simulated runs)
O5 Resources used or wasted	_____

Notes and anomalies

Anything unusual: a dose that differs from other runs, a claim not supported by the transcript, a reply that broke off. Give the turn or time.

Rater signature _____ **Second rater (if any)** _____